Asymptotics and Heteroskedasticity

Lecture 5 (R code)

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Reading

This lecture covers material from Wooldridge (2025):

- Chapter 5.1 - "Consistency"
- Chapter 5.2 - "Asymptotic Normality and Large Sample Inference"
- Chapter 5.3 - "Asymptotic Efficiency of OLS"
- Chapter 8.1 - "Consequences of Heteroskedasticity for OLS"
- Chapter 8.2 - "Heteroskedasticity-Robust Inference after OLS Estimation"
- Chapter 8.3 - "Testing for Heteroskedasticity"

Note: we will not cover 5.4 ("Testing Normality of the Error Term") and 8.2a ("Computing Heteroskedasticity-Robust LM Tests").

Exact properties

The properties of OLS from Lecture 3 and Lecture 4 all hold for the **exact** sample

- *exact*: for a given, finite n
- also referred to as *finite* or *small* sample properties

Begs the question:

- does anything change as n increases?
- what happens in the limit as $n \to \infty$?

Asymptotic properties

Today we will discuss the *asymptotic properties* of OLS

- also referred to as **large sample** properties

Noteably,

1. Consistency
2. Asymptotic normality

CLRM

We will see that as n gets large, we will be able to relax certain assumptions.

1. Replace MLR 4 ($E[u|\mathbf{x}] = 0$) with a weaker uncorrelatedness: $E[u\mathbf{x}] = 0$
2. Relax MLR 6 ($u|\mathbf{x} \sim N(0, \sigma^2)$) entirely

In the second half of the lecture, we will look at heteroskedasticity

3. Relax MLR 5 ($Var(u|\mathbf{x}) = \sigma^2$)

Consistency

In Lecture 3 we solved for,

$$\hat{\beta}_1 = \beta_1 + \underbrace{\frac{1}{\text{SST}_x} \sum_{i=1}^{n} (x_i - \bar{x})}_{\text{all depends on } X} u_i$$

where,

$$\text{SST}_x = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

This was for the *simple linear model*.

We can re-write this as,

$$\hat{\beta}_1 = \beta_1 + \frac{\sum_{i=1}^{n}(x_i - \bar{x})u_i}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \beta_1 + \frac{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})u_i}{n^{-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Then, as $n \to \infty$

$$\hat{\beta}_1 \to_p \beta_1 + \frac{Cov(x_i, u_i)}{Var(x_i)}$$

Probability limit

We say that the probability limit of the OLS estimator is:

$$\text{plim}\ \hat{\beta}_1 = \beta_1 + \frac{Cov(x_i, u_i)}{Var(x_i)}$$

So long as,

$$Var(x_i) \neq 0$$

Which is true under MLR 3 (eqiv. SLR 3).

Proof

Why? This is an application of the **Weak Law of Large Numbers**.[1]

Which holds, so long as the terms have a finite variance.

- $Var(x_i) < \infty$
- $Var(u_i) < \infty$

> **!** Weak Law of Large Numbers
>
> The WLLN states that the sample mean converges in probability to the expected value of a random variable under iid sampling:

[1] It also makes use of Slutzky's theorem which states that if $X_n \to_p X$ and $Y_n \to_p Y$, then $X_n/Y_n \to_p X/Y$; so long as $Y \neq 0$.

$$\bar{x}_n = n^{-1} \sum_{i=1}^{n} x_i \to_p E[x_i]$$

as $n \to \infty$. We typically assume that the variable has a finite variance; though, it is not strictly necessary.

So,

$$n^{-1} \sum_{i=1}^{n} (x_i - \bar{x}) u_i \to_p Cov(x_i, u_i)$$

and,

$$n^{-1} \sum_{i=1}^{n} (x_i - \bar{x})^2 \to_p Var(x_i)$$

Any by Slutzky's theorem, the ratio of the two terms converges in probability too.

Assuming $Cov(x_i, u_i) = 0$ (*weaker than* MLR 4)

$$\text{plim } \hat{\beta}_1 = \beta_1$$

$\to$ the OLS estimator is **consistent**.

Probability limit

The formal definition of consistency is given by,

$$Pr(|\hat{\beta}_1 - \beta_1| > \varepsilon) \to 0 \quad \text{as} \quad n \to \infty$$

for any positive value ε. There is also where the phrase "probability limit" comes from. If you fix ε, then there is a sufficiently large n such that the probability that the estimator is ε away from the true value is close to 0.
We say then that $\hat{\beta}_1$ converges **in probability** to β_1: $\hat{\beta}_1 \to_p \beta_1$.

Consistency

So, under MLR 1-3 PLUS[2]

> **MLR-4a: Zero mean and Zero Correlation**
>
> $E[u] = 0$ and $E[ux_j] = 0$ for $j = 1, \dots, k$.

OLS is consistent.

> **!** Important
>
> Consistent is not the same as unbiased. Unbiasedness is a property for an exact n, while constistency concerns the probability limit of the estimator as $n \to \infty$. Estimators can be consistent, but biased. In some instances, estimators can be unbiased but not consistent because their variances do not shrink as n increases.

MLR 4 vs MLR 4a

What is difference? Under MLR 4, the population regression function is correctly specified

$$E[y_i|\mathbf{x}] = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$$

Under MLR 4a this need not hold.

- the model may be missing some non-linear functions of x_j: terms in x_j^2, x_j^3, etc.
- the model may be missing interactions of regressors.

Simulation

We can demonstrate consistency by showing that as n increases, the variance of the estimator collapses around the true value.

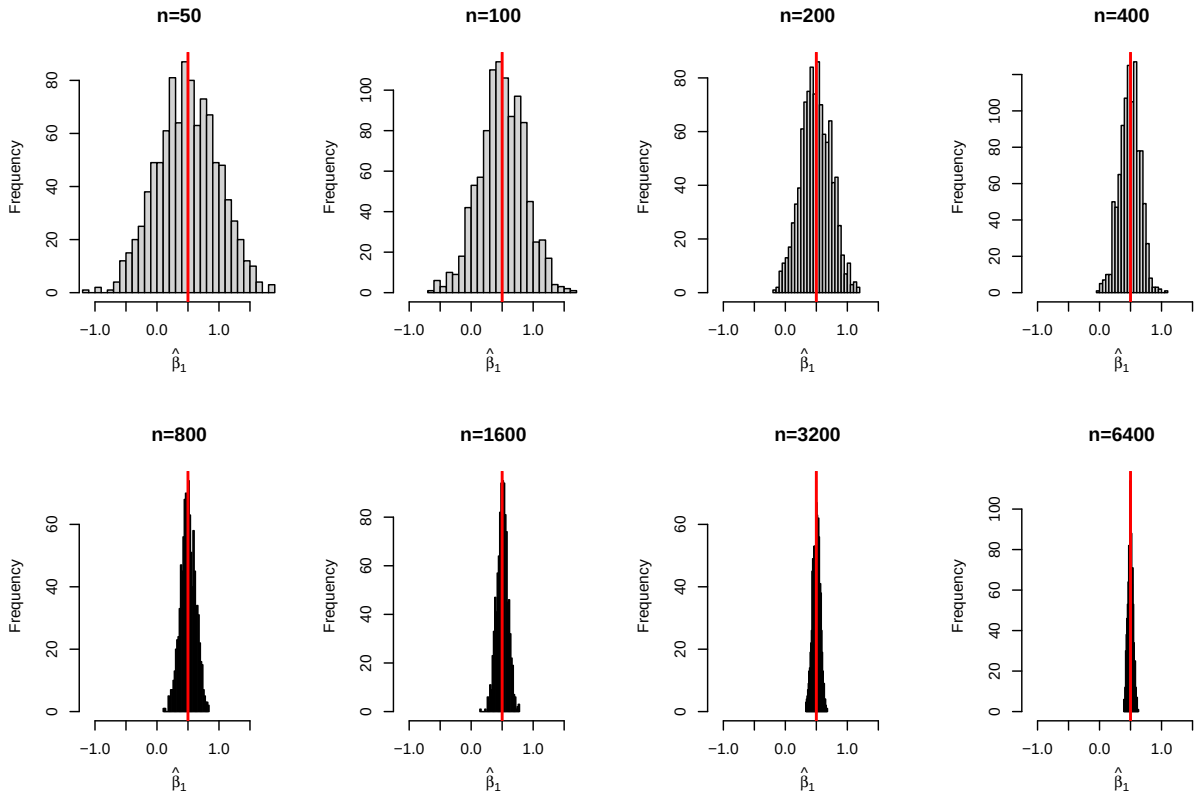
```
Mean of slope estimates:
```

```
  n=50   n=100   n=200   n=400   n=800 n=1600 n=3200 n=6400
0.4915 0.4984 0.4933 0.4892 0.5026 0.5034 0.4997 0.4981
```

```
SD of slope estimates:
```

[2]Here I use the label MLR 4a ('a' for alternative). Wooldridge uses 4', which can get lost in notation. Both are acceptable.

```
  n=50   n=100   n=200   n=400   n=800 n=1600 n=3200 n=6400
0.4888 0.3646 0.2409 0.1633 0.1198 0.0886 0.0599 0.0419
```



Multivariate case

Is anything different in the multivariate case? **No**

We know that we can define the OLS estimator for β_1 from,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

As a simple linear regression of residualized variables $\tilde{y}$ and $\tilde{x}$:

$$\tilde{y}_i = \beta_0 + \beta_1 \tilde{x}_{i1} + v_i$$

So,

$$\text{plim } \hat{\beta}_1 = \beta_1 + \frac{Cov(x_{i1}^*, u_i)}{Var(x_{i1}^*)}$$

where $x_{i1}^* = \text{plim } \tilde{x}_{i1}$.

Under, MLR 4a

$$Cov(x_{i1}^*, u_i) = 0$$

since x_{i1}^* is just a linear function of $x_{i1}, \dots, x_{ik}$; all of which are uncorrelated with u_i.

Asymptotic Normality

Result

$$\sqrt{n}(\hat{\beta}_j - \beta_j) \to_d N(0, Var(\hat{\beta}_j))$$

where,

$$Var(\hat{\beta}_j) = \text{plim } \frac{\sigma^2}{n^{-1} SSR_j}$$

> **Note**
>
> This formula for limiting variance has an n^{-1} in the denominator that doesn't appear in the exact sample formula for the conditional variance. Wooldridge (2025, 174) includes the formula:
>
> $$n^{-1} SSR_j = n^{-1} \sum_{i=1}^{n} \hat{r}_{ij}^2$$
>
> where $\hat{r}_{ij}$ is the residual from a regression of x_j on all other regressors; *including a constant.*

Why this strange form? With the $\sqrt{n}$.

1. $\hat{\beta}_j \to_p \beta_j$
2. In the limit, the estimator has no distribution (variance=0); degenerate
3. The speed of convergence is $n^{-1/2}$[3]
4. If we scale the estimator (minus it's limit (β_j)) we end up with a non-degenerate distribution
5. This is why SEs of estimator shrinks at roughly $n^{-1/2}$: if you quadruple the sample size, the SE halves.

BUT, if you de-mean $\hat{\beta}_j$ and scale by $\sqrt{n}$, the term converges to a stable, **normal** distribution.

Proof

The result is a direct application of the **Central Limit Theorem**

- $\sqrt{n}$ (root-n) is the scaling applied to a series of random variables to make them converge to a normal distribution.

> **!** Central Limit Theorem
>
> CLT states that given a **sequence** of iid random variables $x_1, x_2, ..., x_n$, such $E[x_j] = 0$ and $Var(x_j) = \sigma^2 < \infty$.
>
> $$\sqrt{n}\bar{x}_n = n^{-1/2} \sum_{i=1}^{n} x_i \to_d N(0, \sigma^2)$$
>
> as $n \to \infty$.

Standardization

The result can be standardized,

[3] We say OLS is $\sqrt{n}$-consistent (or "root-n" consistent)

$$\begin{aligned}\frac{\sqrt{n}(\hat{\beta}_j - \beta_j)}{\sqrt{\sigma^2/(n^{-1}SSR_j)}} &= \frac{\sqrt{n}(\hat{\beta}_j - \beta_j)}{\sqrt{n}\sqrt{\sigma^2/SSR_j}} \\ &= \frac{(\hat{\beta}_j - \beta_j)}{sd(\hat{\beta}_j)} \\ &\to_d N(0,1)\end{aligned}$$

Under MLR 1-5; **even without MLR 6**

Approximate distribution

We say that,

$$\frac{(\hat{\beta}_j - \beta_j)}{sd(\hat{\beta}_j)} \overset{a}{\sim} N(0,1)$$

where $\overset{a}{\sim}$ means "approximately distributed".

- for a large enough n, it does not matter that you do not know the distribution of $u|\mathbf{x}$.

Estimated variance

Likewise,

$$\frac{(\hat{\beta}_j - \beta_j)}{se(\hat{\beta}_j)} \to_d N(0,1)$$

where $se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2/SSR_j}$

- for a large enough n, it does not matter that you do not know σ^2 either.

t-distribution

As was suggested in Lecture 3,

$$t_{n-k-1} \to N(0,1) \quad \text{as} \quad n \to \infty$$

BUT remember

- t distribution result relies on MLR 1-6.
- MLR 1-5 gives you normality, by CLT.

F-distribution

Also under MLR 1-6, we had a range of tests that had a F distribution (see Lecture-4)

$$F_{q,n-k-1} \to \chi^2_q/q \quad \text{as} \quad n \to \infty$$

As above,

- F distribution of linear tests relies on MLR 1-6.
- MLR 1-5 gives you chi-squared, by CLT.
- Notice, $q \times F_{q,n-k-1} \to \chi^2_q \quad \text{as} \quad n \to \infty$

t and F distribution limits

The t and F distributions share a common denominator:

$$t_{n-k-1} = \frac{z}{\chi^2_{n-k-1}/(n-k-1)} \quad \text{and} \quad F_{q,n-k-1} = \frac{\chi^2_q/q}{\chi^2_{n-k-1}/(n-k-1)}$$

The distribution in the denominator arises from the variance estimator, which has a $\chi^2_{n-k-1}/(n-k-1)$ (conditional) distribution (under MLR 1-6).
As $n \to \infty$,

$$\chi^2_{n-k-1}/(n-k-1) \to 1$$

And in neither cases does the numerator depend on n. This explains the results above.

Hypothesis tests

Having said that, it is equally accurate to say

$$t\text{-stat} \overset{a}{\sim} t_{n-k-1}$$

and

$$F\text{-stat} \overset{a}{\sim} F_{q,n-k-1}$$

without MLR 6.

- Likewise for Chow test.

Demonstration

Let's simulate a model:

$$y_i = 2 + 0.5x_i + u_i$$

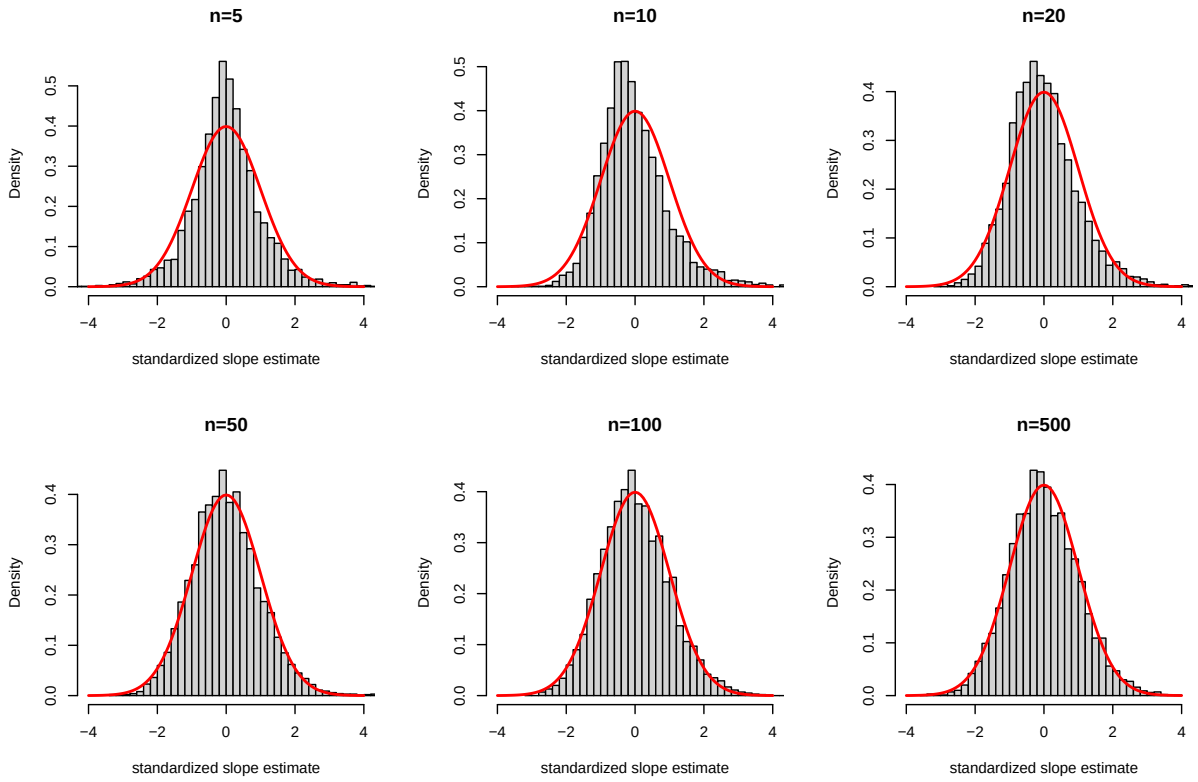
where,

$$u_i + 1 | x_i \sim \text{Exp}(1)$$

- $\Rightarrow E[u_i|x_i = 0]$
- $\Rightarrow Var(u_i|x_i) = 1$
- BUT the error has a **skewness** factor of 2.

Simulation

Here we plot the standardized coefficients.



Asymptotic Tests

The Lagrange Multiplier (LM) statistic

> **Note**
>
> We are covering this because it is relevant to tests for Heteroskedasticity (Ch. 8.3).

The LM-stat can be used under MLR 1-5 (therefore MLR 6 is not needed).

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{k-q} x_{i,k-q} + \beta_{k-q+1} x_{i,k-q+1} + \cdots + \beta_k x_{ik} + u_i$$

Suppose, you want to test

$$H_0 : \beta_{k-q+1} = 0, \ldots, \beta_k = 0$$

- The last q parameters are all 0.

- We saw a version of this for the F-test

LM-stat

You can compute the LM-stat as follows:

1. estimate the **restricted** model

$$y_i = \tilde{\beta}_0 + \tilde{\beta}_1 x_{i1} + \cdots + \tilde{\beta}_{k-q} x_{i,k-q} + \tilde{u}_i$$

where $\tilde{u}_i$ is the residual from this restricted-model regression.

2. regress $\tilde{u}_i$ on $x_1, \ldots, x_k$ (all k regressors)
    - intuitively none of the regressors should be significant under the H_0
3. store the $\mathbf{R}^2_{\tilde{u}}$ from this *auxilliary* regression
4. construct the LM-stat

$$LM\text{-stat} = n \times \mathbf{R}^2_{\tilde{u}}$$

5. Under the null (and assuming MLR 1-5): LM-stat $\to_d \chi^2_q$
6. Compute p-value or use critical value to form rejection rule.

Recap

1. We can relax MLR 4
    - Under MLR 4a OLS is consistent, but potentially biased in the finite sample[4]
2. We can drop MLR 6
    - Regardless the true (conditional) distribution of u_i, the OLS estimator will be approximately normal for n large
    - Conduct inference based on standard normal and chi-squared distributions.

[4] Recall from Lecture 3, unbiasedness requires MLR 4.

- Tests using t/F distribution remain slightly more conservative if distribution is unknown.
- LM test option

3. OLS is asymptotically efficient under Gauss-Markov assumptions (MLR 1-5) (read Wooldridge 2025, ch. 5.3)

Heteroskedasticity

MLR #5

MLR-5: Homoskedasticity

The variance of the error term is the same given any value of $\mathbf{x}$.

$$Var(u|\mathbf{x}) = \sigma^2$$

- States the error variance is the same constant, independent of $\mathbf{x}$

What if this is NOT true?

Heteroskedasticity

A heteroskedastic error variance can be denoted:

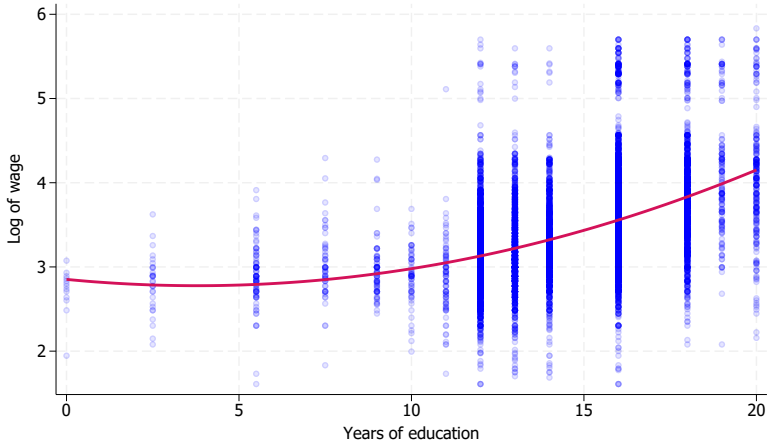
$$Var(u_i|\mathbf{x}_i) = \sigma_i^2$$

- the i shows that the variance depends on i (or the value of $\mathbf{x}_i$)[5]
- implies that the sampling error is NOT iid (independently and *identically* distributed)

[5]Wooldridge (2025, 278) states "variance of the error depends upon the particular value of x_i". This is one form of heteroskedasticity; e.g. $\sigma_i^2 = \sigma^2 x_i$. But heteroskedasticity can be more general than this.

Example

Years of education (modelled here with quadratic term in `edu`; using MORG 2024).



Implications

From Lecture 3, for the *simple linear regression* case,

$$Var(\hat{\beta}_1|X) = E\left[\left(\frac{1}{\text{SST}_x}\sum_{i=1}^{n}(x_i - \bar{x})u_i\right)^2 \middle| X\right]$$

To get the result from Lecture 3 we used MLR 5, **which no longer holds**.

Variance

$$Var(\hat{\beta}_1|X) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2 E[u_i^2|X]}{\text{SST}_x^2} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sigma_i^2}{\text{SST}_x^2}$$

The formula is very similar to what you have under homoskedasticity,

$$Var(\hat{\beta}_1|X) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sigma^2}{\text{SST}_x^2} = \frac{\sigma^2}{\text{SST}_x}$$

except it cannot be simplified, because σ_i^2 depends on i.

Estimator

The most intuitive estimator for the variance is given by,

$$\widehat{Var}(\hat{\beta}_1|X) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2\hat{u}_i^2}{\text{SST}_x^2}$$

as it replaces the error term with the estimated residual.

- Referred to as the White (1980) estimator.
- The estimator has the appropriate asympotitic limit.

Multivariate case

In the multivariate case,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + u_i$$

we had that under **homoskedasticity**,

$$Var(\hat{\beta}_j|X) = \frac{\sigma^2}{\text{SSR}_j}$$

- where SSR_j is from a regression of regressor x_j on all other regressors.

Under **heteroskedasticity**, we

$$Var(\hat{\beta}_j|X) = \frac{\sum_{i=1}^{n}(\tilde{x}_{ij} - \bar{\tilde{x}}_j)^2\sigma_i^2}{\text{SSR}_j^2}$$

where $\tilde{x}_{ij}$ is the residual from the regression of x_j on all other regressors.

Estimator

The following formula appears in Wooldridge (2025, 279) for the estimator:

$$\widehat{Var}(\hat{\beta}_j|X) = \frac{\sum_{i=1}^{n} \hat{r}_{ij}^2 \hat{u}_i^2}{\text{SSR}_j^2}$$

- $\hat{r}_{ij}$ is the residual from a regression of x_j on all other regressors **as well as a constant**.
  - $\sum_{i=1}^{n} \hat{r}_{ij} = 0$: the mean of the residual is zero
  - $(\tilde{x}_{ij} - \bar{\tilde{x}}_j)^2$ is replaced with $\hat{r}_{ij}^2$

HC's

Note: this is not the only **heteroskedastic-consistent** (HC) SE estimator.[6]

- **HC0**: Original White (1980) formula that uses the unadjusted squares of the residuals. **Formula from Wooldridge (2025).**
- **HC1**: Huber-White sandwich formula. Correct's HC0 for degrees of freedom: HC0 $\times n/(n-k-1)$. **Stata default.**
- **HC2**: Adjusts residuals by a regressor-specific leverage term. Better for small samples and corresponds to Neyman variance estimator (in simple linear case with dummy regressor).
- **HC3**: The 'jackknife' estimator. Different adjustment for leverage term. Better small-sample bias adjustment and recommended for small sample settings: $n < 250$.

> Robust SEs
>
> Heteroskedasticity is not the only concern related to SEs. As you read applied work you will see corrections for clustering, auto-correlation, spatial correlations, etc. And when you look into the options, you will read about approaches like bootstrap and jackknife SE estimation. For example, MacKinnon and White's (1985) HC3 correction is the algebraic equivalent of a jackknife ("leave-one-out") estimator.

[6] For a detailed discussion, you can read this Stata blog from 2022.

Consequences

1. Does heteroskedasticity make OLS biased?
    - **No!** (Un)biasedness depends on MLR 1-4
    - the estimation of heteroskedastic-robust SEs has no bearing on the estimated OLS coefficients.
    - heteroskedasticity *may* be the result of omitted variables and/or model mis-specification.
2. Does it make it less efficient?
    - **Yes**. Under heteroskedasticity, OLS is no longer BLUE
    - See Weighted Least Squares Estimation (Ch. 8.4, not examinable)
3. t tests
    - exact distribution of t-stats is not t distributed
    - the asymptotic distributions remain normal
    - can use t distributions as an approximation for larger n (slightly more conservative)
4. F tests
    - there is a heteroskedastic-robust F-stat (also referred to Wald Statistic)
    - Chow test is not valid, but can use dummy variable interaction method with robust F test.
    - Can use heteroskedasticity-robust LM test method

Example: MORG (2024)

```r
# Load the Stata file
morg_data <- read_dta(file.path(data_dir, "morg24_all.dta"))

# Packages
library(sandwich)
library(lmtest)

# Estimate bivariate linear regression
model <- lm(lnwage ~ edu + exp + expsq + female + unioncov, data=morg_data)

# Compute HC1 robust SEs
vce = vcovHC(model, type="HC1")

# Display
coeftest(model, vcov = vce)
```

```
t test of coefficients:

               Estimate  Std. Error t value  Pr(>|t|)
(Intercept)  1.3804e+00  8.7910e-02  15.702 < 2.2e-16 ***
edu          1.1262e-01  2.3257e-03  48.423 < 2.2e-16 ***
exp          2.1965e-02  1.3864e-03  15.843 < 2.2e-16 ***
expsq       -3.1483e-04  2.8434e-05 -11.072 < 2.2e-16 ***
female      -2.0267e-01  1.1101e-02 -18.257 < 2.2e-16 ***
unioncov     1.0915e-01  4.0774e-02   2.677  0.007441 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Or do this in one step
#coeftest(model, vcov = vcovHC(model, type="HC1"))
```

Robust SEs

Are heteroskedastic-robust SEs bigger? **Not necessarily**.

- In practice, they can return smaller SEs.
- Implications for power of the test are therefore ambiguous

```
* Use the vce() option with robust
est clear
qui reg lnwage edu exp expsq female unioncov
est store Homoskedastic

qui reg lnwage edu exp expsq female unioncov, vce(robust)
est store HC1_Robust

etable, est(Homoskedastic HC1_Robust) column(estimates) novarlabel cstat(_r_b, nformat(%5.4f)
```

```
                        Homoskedastic HC1_Robust
-----------------------------------------------
edu                            0.1126     0.1126
                              (0.0021)   (0.0023)
exp                            0.0220     0.0220
                              (0.0014)   (0.0014)
expsq                         -0.0003    -0.0003
                              (0.0000)   (0.0000)
female                        -0.2027    -0.2027
                              (0.0111)   (0.0111)
unioncov                       0.1092     0.1092
                              (0.0473)   (0.0408)
_cons                          1.3804     1.3804
                              (0.0997)   (0.0879)
Number of observations            8955       8955
-----------------------------------------------
```

Why bother?

Why bother with homoskedastic errors?

- if the error variance is homoskedastic **and normally distributed** then the exact distribution of t-stat (F-stat) is known for **small sample**.
- no need to use asymptotic approximation
- heteroskedastic-robust SEs (and t-stats) are only valid for larger sample sizes.

Test for heterskedasticity

Can we test whether the data exhibits heteroskedasticity? **Yes**

1. Breusch Pragan test
2. White test (read about this in Ch. 8.3)

Breusch-Pagan test

$$H_0 : Var(u|\mathbf{x}) = \sigma^2$$

Which implies,

$$H_0 : E(u^2|\mathbf{x}) = \sigma^2$$

- We are testing whether the conditional expectation function of u^2 is independent of $\mathbf{x}$.

For a random sample, if[7]

$$u_i^2 = \delta_0 + \delta_1 x_{i1} + \delta_2 x_{i2} + \cdots + \delta_k x_{ik} + \nu_i$$

Then the null is:

$$H_0 : \delta_1 = \delta_2 = \cdots = \delta_k = 0$$

- These are multiple linear hypotheses

We can test this using a F-test of the above regression.

- **However**, we do not observe u_i^2.
- Replace with $\hat{u}_i^2$
- Regress:

$$\hat{u}_i^2 = \delta_0 + \delta_1 x_{i1} + \delta_2 x_{i2} + \cdots + \delta_k x_{ik} + v_i$$

[7]The White test allows for higher order polynomials of x's and interactions, which reflects the fact that we do not the know the exact form of the CEF.

- Compute F-stat or LM-stat[8]

$$F\text{-stat} = \frac{\mathbf{R}^2_{\hat{u}^2}/k}{(1-\mathbf{R}^2_{\hat{u}^2})/(n-k-1)} \quad \text{or} \quad LM\text{-stat} = n \times \mathbf{R}^2_{\hat{u}^2}$$

Example: MORG (2024)

In R, the relevant tests are a part of the `lmtest` package.

Breusch-Pagan test (F-stat) - can't find matching R version.

```
# 1. Extract squared residuals and run auxiliary regression
aux_model <- lm(I(residuals(model)^2) ~ edu + exp + expsq + female + unioncov, data=model.fra

# 2. View F-statistic, degrees of freedom, and p-value
summary(aux_model)
```

```
Call:
lm(formula = I(residuals(model)^2) ~ edu + exp + expsq + female +
    unioncov, data = model.frame(model))

Residuals:
    Min      1Q  Median      3Q     Max
-0.5714 -0.2351 -0.1395  0.0176  5.4707

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.794e-01  1.022e-01  -3.711 0.000208 ***
edu          3.171e-02  2.125e-03  14.921  < 2e-16 ***
exp          5.373e-03  1.455e-03   3.692 0.000224 ***
expsq       -3.406e-05  2.857e-05  -1.192 0.233250
female      -6.764e-02  1.140e-02  -5.933 3.08e-09 ***
unioncov     6.575e-02  4.848e-02   1.356 0.175062
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5338 on 8949 degrees of freedom
Multiple R-squared:  0.03192,   Adjusted R-squared:  0.03138
F-statistic: 59.01 on 5 and 8949 DF,  p-value: < 2.2e-16
```

[8]The name Breusch Pragan test is usually reserved for the LM test equivalent where the test-statistic is $LM = n \times \mathbf{R}^2_{\hat{u}^2}$.

Breusch-Pagan test (LM-stat)

```
# Using lmtest package
bptest(model, studentize = FALSE)
```

```
    Breusch-Pagan test

data:  model
BP = 573.26, df = 5, p-value < 2.2e-16
```

```
# Using olsrr package
#library(olsrr)       # Not available in AppsAnywhere
#ols_test_breusch_pagan(model, rhs=TRUE)

# Using skedastic package
#library(skedastic)   # Not available in AppsAnywhere
#breusch_pagan(model, koenker = FALSE)
```

White test (LM-stat)

```
# LM-statistic: e.g.
library(skedastic)    # Not available in AppsAnywhere
white(model, interactions = TRUE)
```

```
# A tibble: 1 x 5
  statistic  p.value parameter method       alternative
      <dbl>    <dbl>     <dbl> <chr>        <chr>
1      403. 4.97e-73        20 White's Test greater
```

References

Bibliography

Wooldridge, Jeffrey M. 2025. “Introductory Econometrics: A Modern Approach.”